Beta-Divergence-Based Recurrence Analysis of EEG Signals for Seizure Detection

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***Abstract*—Recurrence Plots (RPs) describe the dynamics of time series through proximity relations in reconstructed phase space, typically using Euclidean distance. However, this metric may be inadequate for challenging time series such as electroencephalogram (EEG) signals. In this work, we propose the use of β-divergence as an alternative dissimilarity metric for RP-based seizure detection. Recurrence Quantification Analysis (RQA) is used to generate features to train a Support Vector Machine (SVM) classifier. The results show a strong dependence on the value of β, with several values outperforming the (default) squared Euclidean case ($\beta = 2$). Noise robustness experiments further indicate that specific β values can preserve high accuracy under adverse conditions, highlighting the potential of β-divergences to improve both accuracy and robustness in automated EEG RP-based seizure detection.**

***Index Terms*—Recurrence Plots, Recurrence Quantification Analysis, β-divergence, Electroencephalogram, EEG.**

I. Introduction

Recurrence plots (RPs) have been widely applied in signal processing, including engineering [1], biological [2], and medical applications [3], [4]. By transforming one-dimensional time series into two-dimensional representations, RPs reveal recurrent patterns and temporal structures of the underlying dynamical systems [5]. Their effectiveness for EEG analysis, particularly in capturing seizure-related dynamics, has been demonstrated in several studies [3], [6].

To improve RP-based seizure detection, we investigate the use of β-divergences [7] for RP generation. β-divergence is a parametric family of measures that generalizes several prominent divergences: the squared Euclidean distance ($\beta = 2$), the Kullback–Leibler divergence ($\beta = 1$), valued in information theory for measuring spectral similarity [8], and the Itakura–Saito divergence ($\beta = 0$), noted for its scale-invariant properties [7], [9]. Despite its versatility, the behavior of these divergences in the context of EEG-based RPs has not yet been systematically studied. By tuning the parameter β, the divergence can selectively emphasize discrepancies in different amplitude regimes: lower values accentuate differences among low-amplitude components, whereas higher values increasingly penalize deviations at larger amplitudes [10]. In addition, we introduce a unified thresholding strategy for RP construction, designed to remain consistent across different values of β and to avoid distribution-dependent tuning.

The methodology proposed in this work starts by preprocessing raw EEG signals to remove outliers and normalize amplitudes, followed by decomposition into standard frequency bands: delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), and beta (13–30 Hz). RPs are then constructed for each band via phase-space reconstruction, using heuristics for the time delay and embedding dimension parameters, with β-divergence as the core metric. To evaluate these representations, Recurrence Quantification Analysis (RQA) [5] is applied to extract nonlinear features (e.g., recurrence rate, determinism, etc.), which subsequently serve as inputs for a Support Vector Machine (SVM) [11] classifier aimed at seizure detection.

The main contributions of this work include the systematic exploration of β-divergence for EEG RPs, generalizing the Euclidean case to better capture non-stationary ictal dynamics. To support this, we introduce a unified, distribution-independent thresholding strategy based on the Median Absolute Deviation (MAD). Furthermore, we provide an evaluation of noise robustness, identifying specific β ranges that offer a superior trade-off between accuracy and stability.

The remainder of this paper is structured as follows. Section II describes the methodology, including phase-space reconstruction, the mathematical framework of β-divergence, the novel thresholding technique, and the RQA-SVM classification pipeline. Section III details the experimental setup. Section IV presents performance metrics for seizure detection and noise robustness for a range of β values. Section V provides an analysis of the results, their implications and directions for future work. We conclude in Section VI.

II. Methodology

First, we briefly recall RP construction and β-divergence as a dissimilarity measure, forming the basis of the proposed methodology. The specific processing steps adopted in this work are then detailed in the following subsections.

Let $s(t)$ denote a signal. RPs are constructed via phase-space reconstruction using time-delay embedding. After sampling the signal at times t_i, define $s_i := s(t_i)$. The embedded state vectors are

$$\mathbf{x}_i = \left[s_i,\ s_{i+\tau},\ \ldots,\ s_{i+(m-1)\tau}\right]^\top, \qquad i = 1, \ldots, N, \quad (1)$$

where m is the embedding dimension and τ is the time delay. This yields a collection $\{\mathbf{x}_i\}_{i=1}^N$, where each $\mathbf{x}_i$ represents the reconstructed state at time index i, and N is the total number of embedded states. The recurrence matrix $\mathbf{R} \in \mathbb{R}^{N\times N}$

is obtained by evaluating pairwise dissimilarities between embedded states,

$$R_{i,j} = d(\mathbf{x}_i, \mathbf{x}_j), \quad 1 \leq i, j \leq N. \tag{2}$$

The RP is obtained by thresholding the recurrence matrix,

$$P_{i,j} = \begin{cases} 1, & d(\mathbf{x}_i, \mathbf{x}_j) \leq \varepsilon, \\ 0, & d(\mathbf{x}_i, \mathbf{x}_j) > \varepsilon, \end{cases} \tag{3}$$

where $\varepsilon > 0$ is a distance threshold that determines when two embedded states are considered recurrent.

An illustrative example of the RP construction process is shown in Figure 1, where $d(\cdot, \cdot)$ is the euclidean distance.

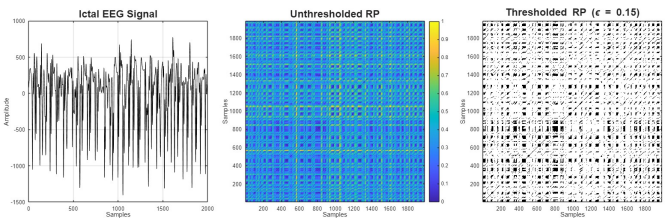

Fig. 1. Example of recurrence plot construction. Left: original ictal EEG signal. Middle: unthresholded recurrence matrix $\mathbf{R}$ obtained with embedding dimension $m = 7$ and time delay $\tau = 2$. Right: thresholded recurrence plot P using $\varepsilon = 0.15$.

In contrast to classical RPs based on Euclidean distance, this work uses β-divergence as the dissimilarity measure $d(\cdot, \cdot)$. The β-divergence is defined, for $x, y > 0$, as

$$d_\beta(x, y) = \begin{cases} \dfrac{x}{y} - \log\left(\dfrac{x}{y}\right) - 1, & \beta = 0, \\ x \log\left(\dfrac{x}{y}\right) + y - x, & \beta = 1, \\ \dfrac{\left(x^\beta + (\beta - 1)y^\beta - \beta x y^{\beta - 1}\right)}{\beta(\beta - 1)}, & \beta \neq 0, 1. \end{cases} \tag{4}$$

When two vectors $\mathbf{x}_i, \mathbf{x}_j \in \mathbb{R}_+^d$ are considered, the β-divergence is computed element-wise:

$$d_\beta(\mathbf{x}_i, \mathbf{x}_j) = \sum_{k=1}^{d} d_\beta(x_{i,k}, x_{j,k}). \tag{5}$$

The following subsections describe the complete processing pipeline, from signal preprocessing and RP construction to feature extraction and classification.

A. Data preprocessing

Data preprocessing consists of outlier removal using Hampel filter [12]. Afterwards the signal is decomposed into standard EEG frequency bands, delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), and beta (13–30 Hz), using a finite impulse response (FIR) filter [13]. A filter order of 200 and a Hamming window are employed to balance frequency resolution and computational efficiency [3]. The normalized signal is shifted to be strictly positive (minimum value = 10^{-7}) to ensure numerical stability and compatibility with β-divergence, which is defined only for positive values.

B. Parameter selection

As stated above, RP construction requires the selection of time delay (τ) and embedding dimension (m), which can significantly impact RP properties. The time delay τ is selected as the first local minimum of the Average Mutual Information (AMI) function [14]. For m, we use False Nearest Neighbors (FNN) [15], selecting the first m for which FNN is less than 1%. Both parameters were selected separately for each frequency band and signal epoch, where a signal epoch corresponds to a 5 seconds EEG segment used to construct an individual RP.

C. Thresholding

For thresholding, i.e., the selection of ϵ, a common approach is to fix a global threshold value [5]. However, this approach is not well-suited for our work because of the differences in distributions of the distances for different β-values and frequency bands. In this work we propose an approach where a specific threshold $\epsilon_{b,\beta}$ is selected for each frequency band b and each β value.

First, in order to ensure that we explore possible threshold values efficiently, the distances in each recurrence matrix $\mathbf{R}_{b,\beta}$ are normalized in the 5th-90th percentile which flattens the distribution. Later, using those normalized distances, thresholds are chosen as:

$$\epsilon_{b,\beta} = x \cdot \mathrm{MAD}(\mathbf{R'}_{b,\beta}), \tag{6}$$

where $\mathrm{MAD}(\mathbf{R'}_{b,\beta})$ denotes the median absolute deviation of the flattened, normalized recurrence matrix $\mathbf{R}'$ for band b and value β, and

$$x \in \Delta, \qquad \Delta = \{k \cdot \delta \mid k = 1, \ldots, N\}, \tag{7}$$

where δ is the step size and N is the total number of thresholds evaluated.

D. Recurrence Quantification Analysis

RQA extracts quantitative features from thresholded RPs to characterize the dynamics of signals [5]. These features reduce the dimensionality of RPs while capturing non-linear and nonstationary behaviors. Standard RQA metrics based on point density, diagonal lines, and vertical lines—specifically Recurrence Rate (RR), Determinism (DET), Average Diagonal Length (L), Entropy (ENTR), Laminarity (LAM), and Trapping Time (TT)—are calculated according to standard definitions found in [5]. However, the use of β-divergence results in asymmetric recurrence plots ($R_{i,j} \neq R_{j,i}$). Unlike standard symmetric RPs, asymmetric RPs require dedicated descriptors for horizontal recurrence structures. Accordingly, in addition to the standard vertical RQA metrics, we consider horizontal laminarity (HOR), average horizontal line length (AH), longest horizontal line (HMAX), and horizontal trapping time type 2 ($T2_H$). In total, 13 RQA metrics are extracted from each thresholded RP. Since the signal is decomposed into four frequency bands (delta, theta, alpha, and beta), a total of 52 features (13 metrics $\times$ 4 bands) are generated for each β value and signal segment.

E. Noise Addition

To assess the robustness of the proposed seizure detection method under varying noise conditions, Gaussian noise was added to the EEG signals. The noise amplitude was scaled relative to the standard deviation of the dataset samples within each class, ensuring realistic noise levels proportional to the signal variability. Specifically, noise levels corresponding to multiples of the standard deviation—ranging from 0 to 0.5 with 0.1 increments—were applied to simulate increasing degrees of contamination.

For reproducibility, the random seed was fixed for each iteration of the cross-validation process. A list of ten distinct seeds, ranging from 1 to 10, was utilized to ensure the stability of the data splitting and training procedures across both classes.

F. Models

An SVM with a Gaussian kernel was employed to classify EEG segments as seizure or non-seizure, using RQA features extracted from the delta, theta, alpha, and beta frequency bands. The model takes a feature matrix of 52 features (13 RQA metrics $\times$ 4 frequency bands) per EEG segment as input, with labels assigned as "Seizure" or "Non-Seizure".

To maintain computational efficiency, hyperparameter optimization was performed on a representative data subset constructed by sampling the lower, median, and upper bounds of the beta and noise parameter ranges. The SVM was trained using the Sequential Minimal Optimization (SMO) solver [16] in MATLAB Bayesian optimization was utilized to tune the box constraint ($C \in [10^{-3}, 10^{4}]$) and kernel scale ($\gamma \in [10^{-3}, 10^{4}]$) within a logarithmic search space, running for 30 iterations with the expected-improvement-plus acquisition function.

III. Experiments

To ensure full reproducibility of the proposed methodology and experimental results, all code used in this study is publicly available in a GitLab repository.[1]

A. Dataset

In this paper, we use the open-access Bonn University EEG dataset [17], which consists of five subsets (A–E), each containing 100 recordings of 23.8 s duration sampled at 173.61 Hz, corresponding to different physiological and pathological conditions in epileptic and healthy subjects. In this work, we focus on subsets D (seizure-free recordings from the epileptogenic zone of epileptic patients) and E (seizure recordings from epileptic patients). These subsets were selected because they reflect the seizure and seizure-free conditions relevant for automated seizure detection, enabling a clinically meaningful binary classification task.

[1]https://github.com/FilipPolakowski/beta_div_recurrence_plots.git

B. Seizure detection

To identify optimal β values for seizure detection, an experiment was designed to evaluate the SVM model's performance across a range of β values and thresholds. The model was trained and tested on a dataset of EEG recordings, using RPs generated with β-divergence as the divergence function. Specifically, 71 β values in the range [0, 7] with an increment of 0.1 were tested, each with 16 threshold coefficients x (see (7)) in the range [0.5, 2].

For each β value, the dataset was partitioned using a stratified split into a training set (80%) and a held-out validation set (20%) across 10 random seeds to ensure robustness. The validation set was strictly used to assess the impact of the β parameter and identify its optimal value, acting as a proxy for generalization performance. The SVM model, described in Section II-F, was trained on the training set's RQA features (13 metrics $\times$ 4 frequency bands) using optimized hyperparameters fixed from a prior search. Five performance metrics, accuracy, sensitivity, specificity, precision, and F1-score, were computed on the validation set for each split. These metrics were averaged across the 10 splits to determine the configuration that maximizes seizure detection performance.

C. Noise robustness

To address noise robustness, the experiment was repeated using EEG samples with added noise, as detailed in Section II-E. Five levels of Gaussian noise were considered, corresponding to standard deviations of 0.1, 0.2, 0.3, 0.4, and 0.5 (excluding the baseline of 0). For each noise level, the same experimental procedure was followed as in the original setup: the dataset was split into 80% training and 20% testing sets across 10 random seeds, and RPs were generated using β-divergence. The SVM model was trained on the RQA features and evaluated using the same five performance metrics.

D. Model explainability

To assess the contribution of each EEG frequency band to the classification model, a band-level permutation importance analysis was performed. Each frequency band was treated as a single feature group by jointly permuting all corresponding RQA features within the test set. For each band, 100 random permutations were applied, and the resulting F1-scores were compared to the baseline performance. This procedure was repeated for each train–test split, and the importance scores were averaged across splits.

IV. Results

A. Seizure detection

Seizure detection performance varies significantly with the choice of the parameter β. Figure 2 presents a comparative analysis of the F1-score across the β spectrum for both noise-free (red) and noisy (blue) conditions.

In the noise-free scenario, the best-performing configuration was obtained for $\beta = 5.2$, achieving an F1-score of 90% on the validation set with a recurrence threshold set to 1.7. This was closely followed by $\beta = 4.9$ (threshold 1.1, F1-score 89.7%).

The standard Euclidean case ($\beta = 2$) served as a reference baseline, yielding a lower F1-score of 86.8% with a threshold of 1.2.

Notably, Figure 2 reveals a distinct shift in the optimal operating point when noise is introduced. Under noisy conditions (averaged standard deviations 0.1–0.5), the peak performance shifts towards lower values. The optimal configuration for the validation set became $\beta = 2.9$ (F1-score 85.9%, threshold 1.9), followed by $\beta = 3.4$. The Euclidean case ($\beta = 2$) remained suboptimal, with an F1-score of 84.7%.

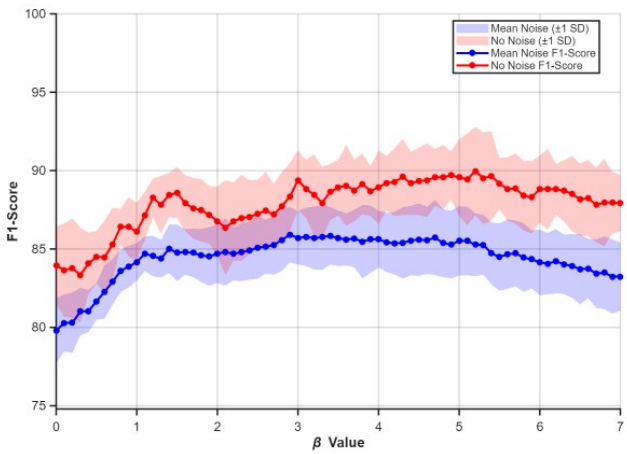

Fig. 2. Performance comparison: Noise-free (red) vs. noisy conditions (blue). The optimal configuration shifts from $\beta = 5.2$ (F1 90.0%) in clean data to $\beta = 2.9$ (F1 85.9%) under noise. In both cases, the Euclidean baseline ($\beta = 2$) remains suboptimal, yielding F1-scores of 86.8% and 84.7%, respectively.

To determine the optimal threshold for each parameter β, multiple threshold values were evaluated during the experiments. The majority of optimal thresholds for the tested β values fell within the range $[1.5, 2]$. Notably, the selected thresholds exhibited consistency between the two experiments, with similar values yielding optimal performance in most cases.

B. Noise robustness

To evaluate the stability of the proposed method, we analyzed the performance degradation induced by additive Gaussian noise. The impact was quantified by calculating the mean absolute change in F1-score relative to the baseline.

Figure 3 summarizes this degradation across the full range of β values. The results reveal a clear trend: noise sensitivity increases as β increases. Lower β values (specifically $\beta \in [0, 2.5]$) exhibit higher robustness, maintaining a relatively low average performance drop of approximately 2% to 2.5%. In contrast, higher β values lead to a more pronounced degradation, with the average F1-score decreasing by up to 5% for $\beta > 6$. This suggests that while higher β values may achieve peak performance in clean signals, they are less stable in the presence of signal artifacts.

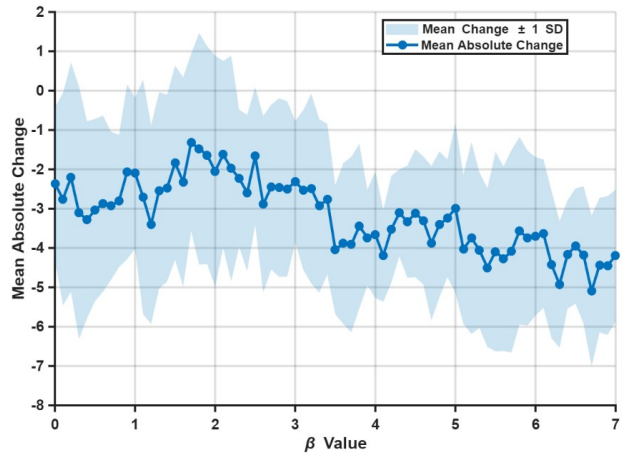

Fig. 3. Average F1-score degradation across all noise levels for each β. Sensitivity increases with β: lower values ($\beta \leq 2.5$) maintain high robustness with smaller drops, whereas higher β values show more pronounced degradation.

C. Model explainability

Figure 4 presents the band-level permutation importance averaged over 100 train–test splits. Importance is measured as the decrease in F1-score after jointly permuting all RQA features within each frequency band. The theta and beta bands exhibit the largest performance reductions.

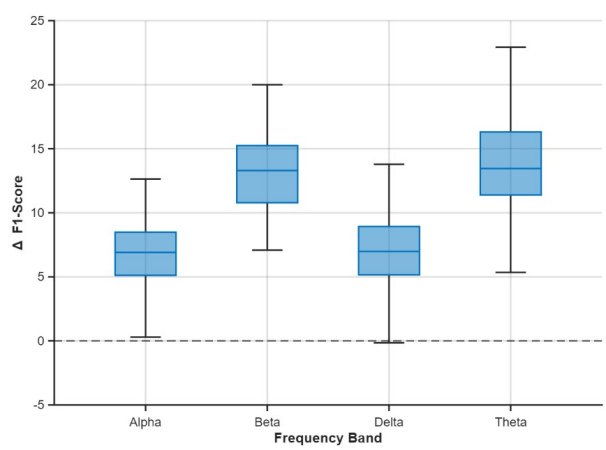

Fig. 4. Band-level permutation importance distribution across 100 train–test splits. Boxplots show the decrease in F1-score after permuting all RQA features within each frequency band.

V. Discussion

The experimental results demonstrate that the choice of the β value in β-divergence RPs can have a substantial impact on seizure detection performance. The poor performance observed for $\beta < 1$ may be attributed to the sensitivity to small signal variations, which are likely amplified by baseline noise present in EEG recordings.

From a practical perspective, the observed F1 score improvement at $\beta \in [3, 5.5]$ over the Euclidean case highlights

the potential of β-divergence-based RPs for enhancing automated seizure detection systems. It is important to emphasize that while a strictly held-out test set would be required to establish the absolute generalization performance for clinical deployment, the primary objective here is to characterize the general trend and sensitivity of the β parameter.

Furthermore, the analysis reveals that robustness to noise is not uniform across the β spectrum. While β values in the higher range (e.g., $\beta > 4$) yield peak performance in clean signals, they exhibit greater sensitivity to noise, as evidenced by larger performance drops. The shift of the optimal configuration towards $\beta \approx 2.9$ in noisy datasets indicates that lower-to-intermediate β values offer a superior trade-off between detection precision and stability against signal artifacts.

It is also important to note that the SVM hyperparameter optimization was conducted on a representative data subset to ensure computational efficiency. While a granular, case-by-case optimization for each specific β value might have yielded marginally superior performance, such an approach would drastically increase the computational cost and the risk of overfitting within the context of a broad exploratory study. However, for a practical application where a small set of optimal β values are selected for deployment, performing a targeted hyperparameter fine-tuning specific to that configuration could be a valuable refinement step.

Finally, the use of β-divergence RPs in combination with an SVM introduces two coupled stages: RP construction and classifier training. Since the quality of the resulting RPs is evaluated indirectly through classification accuracy, it is challenging to disentangle whether performance limitations originate from the representation or from the classifier itself.

VI. Conclusion

This study introduces a β-divergence-based RP approach for seizure detection in EEG signals, addressing the challenges of non-stationarity and noise. Experimental results suggest that optimizing the divergence order could yield significant improvements. Furthermore, the analysis under noisy conditions reveals a trade-off between peak precision and stability: while higher β values excel in clean data, $\beta = 2.9$ proved most robust against signal artifacts, maintaining an F1-score of 85.9%.

The β-divergence-based RP method is a promising direction for automated seizure detection. The observed robustness to noise is particularly relevant in clinical EEG recordings, which are often affected by artifacts and measurement variability.

Building on these results, several directions for future research can be identified. First, the integration of dimensionality-reduction techniques such as PCA could be explored, as they have been shown to enhance classification performance when combined with RQA features [18]. Second, alternative thresholding strategies for RP construction with β-divergence may further improve robustness. Third, given its flexibility in emphasizing different signal amplitude regimes, β-divergence-based RPs could be investigated in other application domains beyond seizure detection. Fourth, combining multiple RPs generated with different β values may allow capturing complementary dynamical information. Fifth, the robustness of the method under common EEG artifacts, such as eye blinks, should be evaluated. Finally, future work could investigate the effect of different noise types and outlier contamination to further characterize the operational limits of specific β ranges in clinical settings.

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